Contraceptive continuity after method discontinuation in Bangladesh

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Stopping a method does not always mean stopping contraception.

A survey-weighted retrospective study of switching, nonuse and reported pregnancy using BDHS 2022.

INTRODUCTION

After a contraceptive method ends, women may switch methods, resume the same method, remain without contraception or report pregnancy. These outcomes answer different questions and need separate measures. [1,2]

OBJECTIVES

Estimate 12-month outcomes after an explicitly defined discontinuation. Assess whether more flexible models improve prediction when entire survey clusters are held out.

METHODS

Data: BDHS 2022 retrospective reproductive calendars. Exploratory secondary analysis; not preregistered. [3]

19,987	Women in the calendar domain
4,516	Complete-horizon eligible discontinuations
1,666	Episodes meeting the reason definition
1,188	Women after selecting earliest eligible episode

Index discontinuation occurred 15-62 months before interview, at ages 15-49, with 12 months of observed follow-up. Reasons indicating pregnancy desire, failure or reduced need were excluded. Retained reasons proxy potential continuing need.

Inference: normalized survey weights, strata and primary sampling units (PSUs); design-based 95% confidence intervals. The primary cohort spans 541 PSUs.

Validation: five repetitions of nested PSU-grouped cross-validation; tuning and transformations fitted within training folds.

RESULTS

20.4% — **Nonuse at month 12, with no prior reported pregnancy**

95% CI 17.8-23.3

12.2% — **Reported pregnancy within 12 months**

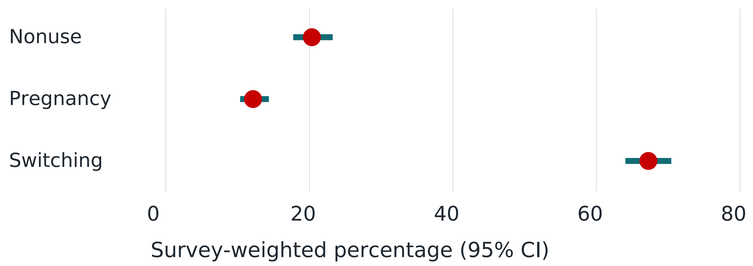
95% CI 10.4-14.4

67.3% — **Different-method switch within 3 months**

95% CI 64.1-70.5

Switching is cumulative and can precede later outcomes. These percentages must not be added. Month-12 nonuse does not mean uninterrupted nonuse for a year.

OUTCOMES WITH UNCERTAINTY



BENCHMARK REPLICATION

Official all-method 12-month discontinuation: **32.2%** published and **32.22%** reproduced. This uses a separate episode population and supports reconstruction, not causal validity. [3]

MODEL PERFORMANCE

Mean history-model ROC AUC across five partitions:

Month-12 nonuse: 0.579-0.601
Reported pregnancy: 0.660-0.679

These ranges compare model means, not confidence intervals. Flexible models produced limited gains. No individual clinical risk tool is established.

SENSITIVITY & INTERPRETATION

A stricter method-related reason definition gave month-12 nonuse of **17.7%** (95% CI 15.1-20.6).

The standardized richest-minus-poorest nonuse difference was **7.2 percentage points** (95% CI -4.0 to 18.3). The data do not establish a clear wealth gradient.

LIMITATIONS

Retrospective monthly reports are subject to recall error. The cohort is selected and does not represent all women in Bangladesh. Interview-time wealth may follow the historical outcome. Recorded pregnancy does not establish intention or clinical confirmation. Associations do not estimate intervention effects.

CONCLUSIONS

Different-method switching was common, while nonuse and reported pregnancy remained relevant outcomes. Transparent definitions and survey-aware uncertainty are essential for interpreting contraceptive continuity.

ACKNOWLEDGEMENTS

Data source: NIPORT and ICF / The DHS Program. The analysis uses the 2022 BDHS reproductive calendars. The authors thank the survey participants and data-collection teams.

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[3] NIPORT & ICF. Bangladesh Demographic and Health Survey 2022: Final Report (2024), Table 7.8.